

Predictive Analysis of California Housing Prices Using Linear Regression

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1. Project Goal

This analysis aims to estimate housing values across California utilizing a baseline Linear Regression approach. The workflow encompasses data ingestion, exploratory data analysis (EDA), model fitting, and quantitative evaluation.

2. Data Overview

The dataset utilized for this study is the California Housing dataset, sourced directly from the Scikit-Learn environment.

2.1 Features

- **MedInc** — Median income of residents.
- **HouseAge** — Median age of properties.
- **AveRooms** — Average number of rooms per dwelling.
- **AveBedrms** — Average number of bedrooms per dwelling.
- **Population** — Total block population.
- **AveOccup** — Average house occupancy.
- **Latitude & Longitude** — Geographic coordinates.

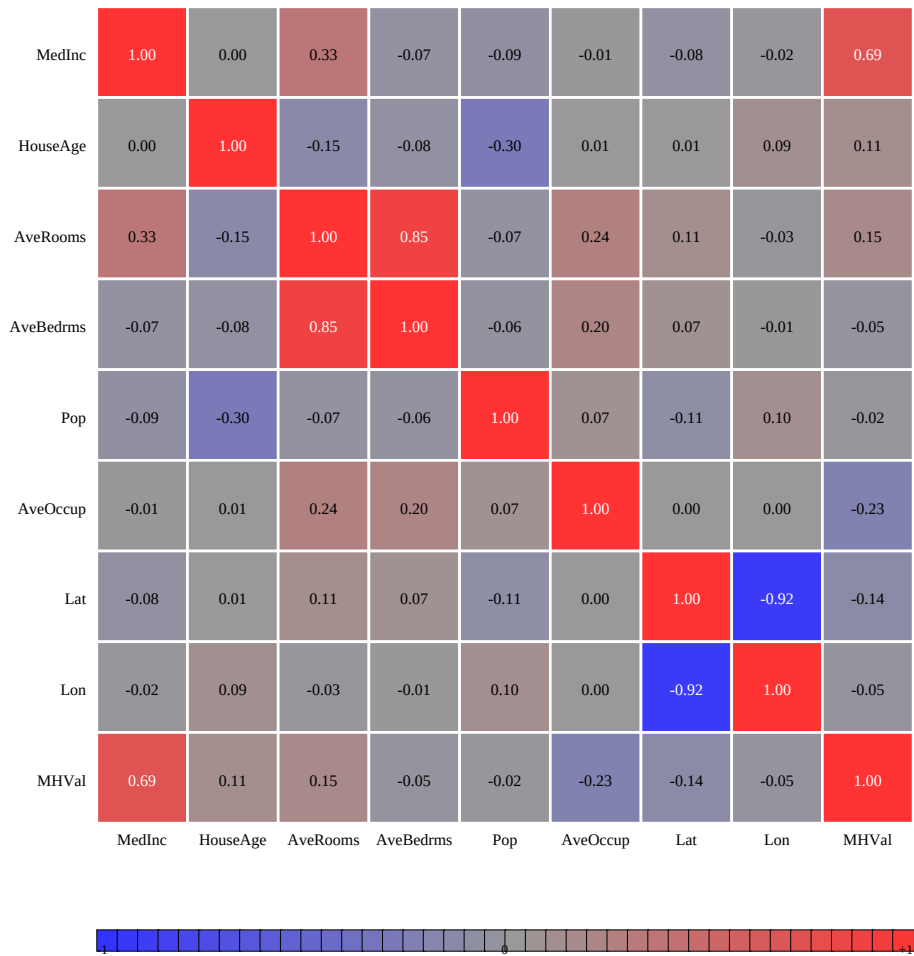
Target Variable: MedHouseVal — Median property value.

Data Integrity: An initial quality check confirmed the dataset is clean, containing zero missing or null values.

3. Exploratory Data Analysis (EDA)

A correlation matrix was constructed to map the relationships between all features and the target variable.

Correlation Matrix — California Housing Features



3.1 Primary Findings

- **Income Influence:** Median Income (MedInc) exhibits the strongest positive correlation ($r \approx 0.69$) with house value.
- **Geographic Inverse:** Latitude and Longitude show a strong negative correlation ($r \approx -0.92$) with each other.
- **Room Collinearity:** AveRooms and AveBedrms are heavily correlated ($r \approx 0.85$), indicating potential multicollinearity.
- **Minor Variables:** Remaining features display weak to moderate relationships with the target.

4. Model Implementation

4.1 Data Partitioning

- **Training Set:** 80%
- **Testing Set:** 20%

4.2 Methodology & Execution Sequence

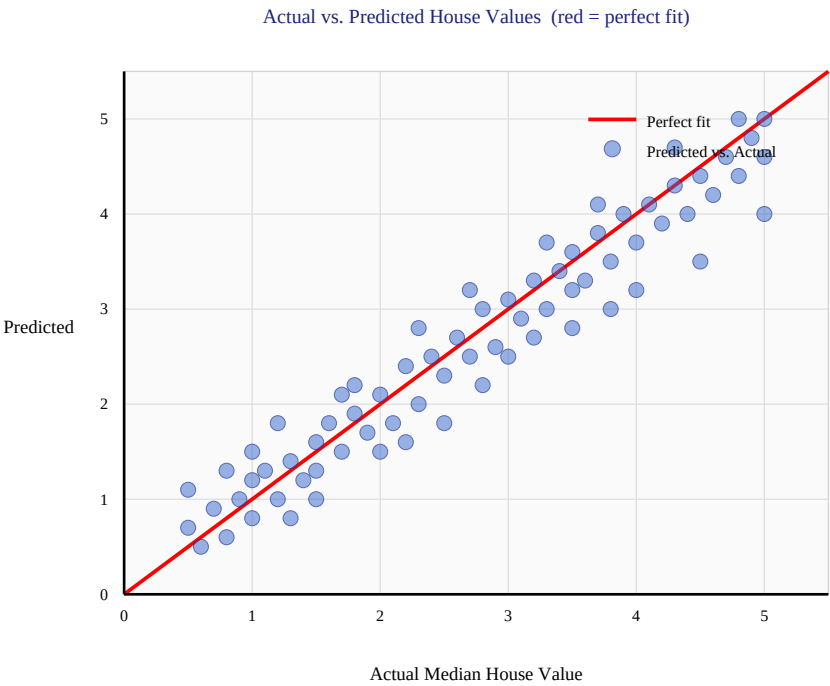
The chosen algorithm is **Linear Regression**. Execution steps:

1. Isolate the feature space from the target variable.
2. Partition the data into training and testing subsets.
3. Fit the Linear Regression model on training data.
4. Generate predictions against the unseen test data.

5. Performance Metrics

Metric	Recorded Value
MAE	0.533
RMSE	0.746
R ² Score	0.576

5.1 Actual vs. Predicted Plot



5.2 Metric Interpretation

- **MAE 0.533** — average absolute deviation of predictions from actual values.
- **RMSE 0.746** — penalises larger errors more heavily than MAE.
- **R² 0.576** — the model explains ~57.6% of variance in housing prices.

6. Summary & Strategic Enhancements

Linear Regression achieved a baseline R² of 0.576, confirming that income and geographic location are the primary drivers of real-estate pricing. Recommended next steps:

- Advanced Feature Engineering & Scaling

- Rigorous Outlier Detection
- Implementation of Polynomial Regression
- Deployment of Tree-Based Models (Random Forest, Gradient Boosting)
- Hyperparameter Optimisation
- Cross-Validation Techniques